

MD ABDUR RAHMAN, PhD

Nanofabrication & Thin-Film Scientist | Optical Coatings | Process Integration

Singapore | abdur82@alumni.kaist.ac.kr | [linkedin.com/in/md-abdur-rahman-3228a443](https://www.linkedin.com/in/md-abdur-rahman-3228a443)

E Pass Holder; Citizenship: Bangladeshi

PROFESSIONAL SUMMARY

Materials scientist and nanofabrication researcher with 10+ years of experience advancing thin-film optical coatings, photonic devices, micro/nanofabrication, and materials characterization. Translates concepts from optical and materials modeling through process development, device fabrication, experimental validation, industry collaboration, and technology transfer. Combines deep technical execution with a sustained record of invention, publication, mentoring, and cross-sector collaboration. Co-inventor on four granted or published patent filings and author/co-author of 20+ peer-reviewed publications.

TECHNICAL EXPERTISE

Thin Films & Optical Coatings	Magnetron sputtering, e-beam evaporation, thermal evaporation, ALD, spin coating, single- and multilayer optical design, structural color
Micro/Nanofabrication	Photolithography, maskless laser lithography, electron-beam lithography, ICP-RIE, wet/dry etching, metallization, lift-off, binary lithography and reflow; Raith eLINE Plus
Metrology & Characterization	Ellipsometry, UV-Vis and microspectrophotometry, Raman, TEM, SEM, AFM, XPS, XRD, EDS, optical and surface profilometry
Modeling, Layout & Analysis	Lumerical, OpenFilters, Python, MATLAB, OriginPro, ImageJ, KLayout

PROFESSIONAL EXPERIENCE

Singapore University of Technology and Design (SUTD)

Senior Research Fellow

Singapore

Jul 2026-Present

- Oversee nanofabrication activities and process strategy for Prof. Joel Yang's multidisciplinary research group within Engineering Product Development.
- Advance sub-10 nm patterning using a novel negative-tone electron-beam resist in collaboration with a resist-development company.
- Currently developing positive-tone PMMA electron-beam lithography processes targeting sub-20 nm features for single-photon-source research in collaboration with RISE Research Institutes of Sweden.
- Provide technical mentoring for thin-film color-coating product development with a Singapore-based company and contribute to industry-oriented government grant proposals.
- Train researchers and industry engineers in cleanroom protocols, process planning, equipment operation, and process troubleshooting.

Research Fellow II

Dec 2024-Jul 2026

- Developed visible and mid-infrared thermoelectric photodetectors using plasmonic and planar device architectures.
- Designed and fabricated photodetector stacks while mentoring doctoral researchers in metallization, lift-off, and device-integration processes.

Research Fellow

Dec 2021-Dec 2024

- Led the nanofabrication workstream in a LiteON Semiconductor-supported program on CMOS-compatible multispectral color filters, producing a peer-reviewed publication and a related U.S. patent application.
- Designed and fabricated 16 distinct Fabry-Perot filters using a single binary-lithography/reflow process with a minimum feature size of 0.6 μm .
- Established a post-CMOS process flow for integrating multispectral Fabry-Perot filters with LiteON-fabricated chips, resolving compatibility constraints and supporting technology transfer.

Korea Institute of Science and Technology (KIST)

Seoul, South Korea

Postdoctoral Researcher

Jul 2017-Nov 2021

- Designed and experimentally realized single- and multilayer thin-film structures for structural-color optical coatings using analytical and numerical modeling.
- Created lithography-free structural-color platforms for optical filters, decorative coatings, and aesthetic surface applications.
- Established precision deposition recipes using magnetron sputtering, thermal evaporation, and spin coating.

- Investigated growth and coalescence mechanisms of epitaxial L10-FePt islands on MgO for heat-assisted magnetic-recording applications.
- Characterized nanoscale structure and perpendicular magnetic domains using high-resolution TEM, XRD, and complementary magnetic and materials metrology.

EDUCATION

PhD, Materials Science and Engineering | KAIST, South Korea, 2017

MPhil, Physics | BUET, Bangladesh, 2010

SELECTED PATENTS AND PATENT APPLICATIONS

- US12292587B2, High-Durability Coloring Metal Member and Method of Producing the Same - granted 2025.
- EP4115764B1, Noble Metal Member Capable of Expressing Vivid Colors and Method of Coloring Its Surface - granted 2025.
- US20240219708A1, Multiple Fabry-Perot Filters and Multispectral Filter Arrays - published 2024; pending.
- WO2025188243A1, Thermoelectric Photodetector Device - published 2025; pending.

SELECTED PUBLICATIONS

- Rahman et al., Scaling Up Multispectral Color Filters with Binary Lithography and Reflow (BLR), Nanophotonics 13, 3671-3677 (2024) - first author.
- Vu and Rahman et al., Tunable Multiresonant Microcavity Exciton-Polaritons in Colloidal Quantum Wells, Nano Letters 25, 6109-6116 (2025) - equal contribution.
- Rahman and Burak et al., Industrial-Grade Structural Coloration with Robust and Vivid TiON-Based Structures, Advanced Optical Materials 14, e03135 (2026) - equal contribution.
- Pan et al., 3D Printed Designer Color Routers with Low Refractive Index for Low-Light Imaging, Nature Communications (2026) - co-author.
- Zhang et al., Nanoscale 3D Printing of Glass for Nanophotonics, Science Advances 11, eadv0267 (2025) - co-author.
- Mori et al., Pick-and-Place Process for Uniform Shrinking of 3D Printed Micro- and Nano-Architected Materials, Nature Communications 14, 5876 (2023) - co-author.

LEADERSHIP AND PROFESSIONAL SERVICE

Nanofabrication facility training and mentoring for academic researchers, doctoral candidates, and industry engineers; technical strategy for multidisciplinary, industry-oriented, and government-funded research.

AWARDS AND HONORS

- Long Service Award (5 Years), SUTD, 2026
- Best Poster Award, ICMAT, 2023
- Best Poster Award (Co-author), Asia Light Conference, 2023
- NSICT Fellowship, Bangladesh
- Fully Funded Doctoral Scholarship, KAIST, 2012-2017